Answer all questions in this section in the spaces provided.

1 A cyclist is testing a new aerodynamic helmet by measuring the drag force experienced while riding at a steady speed. The drag force is measured using a force sensor attached to the helmet.

The drag force is given by the equation

$$F = \frac{1}{2} k \rho A v^2$$

where k is a unitless constant, ρ is the air density, A is the cross-sectional area, and v is the speed of the cyclist.

(a) Using SI base units, show that the equation is homogenous.

[3]

(b) In an experiment, the cyclist measures the quantity of Ft, where t is the time taken for the experiment. The cyclist obtains slightly different measurements each time.

State two possible random errors in this context.

1.....

2. .....

..... [2]

(c) The force sensor has a zero error.

State whether this error affects the accuracy or the precision of the measurements and suggest how the error can be corrected.

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..... [2]